

Ajani Martin
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EDUCATION

University of Maryland
MS Mechanical Engineering

College Park, MD
Class of 2028

Lafayette College
BS Mechanical Engineering

Easton, PA
Class of 2026

HONORS

POSSE Foundation Full Tuition Scholarship

December 2022 – May 2026

- A small, diverse group of 10 talented students carefully selected and trained, to serve as a catalyst for individual and community development

Excel Scholar

August 2023 – May 2026

- Research scholarship awarded to students allowing for intellectual opportunities that fuel students' curiosity and passion for topics that interest them

RELEVANT COURSE WORK

Aerodynamics and Compressible Flow, Engineering Gas Turbines, Engineering Thermal/Fluids, Engineering Thermodynamics, Fluid Mechanics, Dynamics of Physical Systems, Manufacturing and Design, Nature of Engineering Materials

TECHNICAL PROJECTS

EXCEL Research Scholar (Lafayette College Propulsion & Hypersonic Research Laboratory)

August 2023 – May 2026

- Student-driven, air breathing engine propulsion research on combustion and flame stabilization in supersonic airflow using an aft wall fueled cavity flame holder
- Designed and manufactured a cavity flame holder. A component that sits inside of a Scramjet (Supersonic Combustion Ramjet) whose role is to stabilize the combustion process within the hypersonic/supersonic air flow within the engine
- Conducted research to test the most efficient and suitable location within the cavity to introduce a combustible gas (Ethylene) to the supersonic air flow, allowing for continuous controlled combustion at hypersonic/supersonic speeds
- Conducted tests in the Lafayette College Expansion Tube, a device that allows for controlled testing of vehicles that travel above the speed of sound, through the use of high pressure and combustible gasses

ME 475: Thermal / Fluids Systems

January 2025 – May 2026

- Designed and conducted experiments based on the concepts of thermodynamics, fluid mechanics, and heat transfer
- Analyzed experimental results and communicated findings through detailed scientific reports

Independent Study (ANSYS Fluent and Computational Fluid Dynamics Simulation: CFD)

January 2025 – May 2025

- Utilized ANSYS Workbench and Fusion 360 CAD to create scale models to then test through CFD simulation
- Utilized ANSYS Fluent to simulate fluid flow and formation of boundary layers around rocket stability systems, as well as the laminar and turbulent flows around various objects through Computational Fluid Dynamics simulation

WORK EXPERIENCE

Lafayette College Student Researcher | Lafayette College | Easton PA

August 2023 – May 2026

Hours Per Week: 30

- Maintained and operated hypersonics laboratory equipment while leading fellow students through combustion flow and jet propulsion experiments, ensuring safe procedure and proper equipment use

LEADERSHIP EXPERIENCE

ABC (Association of Black Collegians)

Treasurer: Created and managed a budget of \$10,000, that was used to hold organization events

August 2024 – May 2025

Secretary: Organized all organizational events, meetings, and outreach through detailed scheduling

August 2023 – May 2024

NSBE (National Society of Black Engineers)

August 2022 – May 2026

- *Member:* Participate in workshops related to networking, career building, and community building

LIMS (Lafayette Interdisciplinary Music Society)

President

January 2025 – May 2026

- Reestablished a Lafayette College organization to give students an outlet to express their artistic and musical talents unrestricted by musical taste, ethnicity, religious status, or sexuality

SKILLS

Autodesk Fusion 360 CAD, SolidWorks, ANSYS, MATLAB, Arduino, LaVision Davis, Microsoft Excel, Microsoft Word